GAUTHAM KRISHNA GUDUR

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Research Interests

I am interested in building data-efficient, data-centric, and human-centered machine learning systems.

- **Focus Areas:** Foundation models, LLMs, pre-training and post-training, {continual, active, few-shot} learning, token- and parameter-efficient learning (PEFT), reasoning, agentic AI, modality gaps, cross-modal alignment, personalization, adaptation.
- **Applications:** Multimodal and multisensory learning (wearable, physiological, time-series, acoustic), health AI, ubiquitous computing.

Education

Ph.D. in Electrical and Computer Engineering — Aug. 2023 – Present
The University of Texas at Austin — Austin, TX, USA
Advisors: Prof. Joydeep Ghosh (previously Prof. Edison Thomaz) — GPA: 3.9/4.0

B.Tech. in Information Technology — 2013 – 2017
Anna University — Chennai, India

Publications

Author's name appears in **bold**; * denotes equal contribution and joint lead authorship.
Google Scholar: 411 citations · h-index 10 · i10-index 10

Conference, Journal, and Workshop Papers

[1] **Latent-LoRA: Compact Latent-Space Adapters with Gradient-Free Routing for Continual Learning**
RR Azghan, **Gautham Krishna Gudur**, P Turaga, G Pedrielli, H Ghasemzadeh
Findings of EMNLP 2026

[2] **RAST: Resolution-Aware Privileged Structure Transfer for Low-Resolution Audio Activity Recognition**
JH Park, **Gautham Krishna Gudur**, Y Shen, D Liang, E Thomaz
HumanSys Workshop at ACM MobiCom 2026

[3] **WellComp 2026: Ninth International Workshop on Computing and Software Systems for Well-Being**
Gautham Krishna Gudur, H Yoon, YY Wu, T Xia, H Jia
ACM UbiComp Companion 2026

[4] **PCL: Partitioned Continual Learning via Unsupervised Latent Experts for Audio Classification**
Gautham Krishna Gudur*, M Malu*, T Khandait, RR Azghan, A Rayas, P Turaga, J Ghosh, et al.
Machine Learning for Audio Workshop at ICML 2026

[5] **Gated Adaptation for Continual Learning in Human Activity Recognition**
RR Azghan, **Gautham Krishna Gudur**, M Malu, E Thomaz, G Pedrielli, P Turaga, H Ghasemzadeh
IEEE Internet of Things Journal 2026

[6] **AttenGluco: Multimodal Transformer-Based Blood Glucose Forecasting on AI-READI Dataset**

E Farahmand, RR Azghan, NT Chatrudi, E Kim, **Gautham Krishna Gudur**, E Thomaz, G Pedrielli, P Turaga, H Ghasemzadeh
IEEE EMBC 2025

[7] **SVFT: Parameter-Efficient Fine-Tuning with Singular Vectors**
V Lingam*, A Tejaswi*, A Vavre*, A Shetty*, **Gautham Krishna Gudur***, J Ghosh, A Dimakis, E Choi, A Bojchevski, S Sanghavi
NeurIPS 2024
Abridged versions: FITML Workshop at NeurIPS 2024; WANT [Oral Presentation] and ES-FoMo Workshops at ICML 2024.

[8] **Dataset Distillation for Audio Classification: A Data-Efficient Alternative to Active Learning**
Gautham Krishna Gudur, E Thomaz
ENLSP Workshop at NeurIPS 2024

[9] **Can Calibration Improve Sample Prioritization?**
G Tata*, **Gautham Krishna Gudur***, G Chennupati, ME Khan
HILL and HITY Workshops at NeurIPS 2022

[10] **Data-Efficient Automatic Model Selection in Unsupervised Anomaly Detection**
Gautham Krishna Gudur, R Raaghul, K Adithya, S Vasudevan
IEEE ICMLA 2022

[11] **Zero-Shot Federated Learning with New Classes for Audio**
Gautham Krishna Gudur, SK Perepu
INTERSPEECH 2021
Abridged versions: DPML and HAET Workshops at ICLR 2021.

[12] **Resource-Constrained Federated Learning with Heterogeneous Labels and Models for Human Activity Recognition**
Gautham Krishna Gudur, SK Perepu
DL-HAR Workshop at IJCAI-PRICAI, AIoT Workshop at KDD, and MLMH Workshop at NeurIPS 2020

[13] **Bayesian Active Learning for Wearable Stress and Affect Detection**
A Ragav*, **Gautham Krishna Gudur***
MLMH Workshop at NeurIPS 2020

[14] **A Vision-based Deep On-Device Intelligent Bus Stop Recognition System**
Gautham Krishna Gudur, A Ramesh, R Srinivasan
PURBA Workshop at ACM UbiComp 2019

[15] **ActiveHARNet: Towards On-Device Deep Bayesian Active Learning for Human Activity Recognition**
Gautham Krishna Gudur, P Sundaramoorthy, V Umaashankar
EMDL Workshop at ACM MobiSys 2019

[16] **HARNet: Towards On-Device Incremental Learning using Deep Ensembles on Constrained Devices**
P Sundaramoorthy, **Gautham Krishna Gudur**, MR Moorthy, RN Bhandari, V Vijayaraghavan
EMDL Workshop at ACM MobiSys 2018

Preprints

[17] **Counterfactual Self-Evolving Agents for Evidence-Grounded Reasoning**

X Han, Y Wang, C Chen, W Dai, **Gautham Krishna Gudur**, S Li, HH Chung, GD Hager, J Ghosh, P Liang, S Saria
arXiv preprint 2026

[18] **SensorHarness: Optimizing How LLMs Sense the World**
H Yoon, S Im, P Langer, T Lee, **Gautham Krishna Gudur**, J Kim, X Han, YY Wu, N Zumarraga, Y Zhang, R Müller, J Ghosh, Y Yang, SJ Lee
Preprint 2026

[19] **CLAD-Net: Continual Activity Recognition in Multi-Sensor Wearable Systems**
RR Azghan, **Gautham Krishna Gudur**, M Malu, E Thomaz, G Pedrielli, P Turaga, H Ghasemzadeh
arXiv preprint 2025

[20] **GluMind: Multimodal Parallel Attention and Knowledge Retention for Robust Cross-Population Blood Glucose Forecasting Systems**
E Farahmand, RR Azghan, NT Chatrudi, VYA Baidoo, E Kim, **Gautham Krishna Gudur**, et al.
arXiv preprint 2025

Research and Work Experience

The University of Texas at Austin – Graduate Research Assistant — Austin, TX
Advisors: Prof. Joydeep Ghosh, Prof. Edison Thomaz — Aug. 2023 – Present

- Devised test-time LLM user personalization for multimodal time-series health tasks using few-shot in-context learning.
- Proposed SVFT (Singular Vectors guided Fine-Tuning), a Pareto-optimal PEFT technique [Mentor: Prof. Sujay Sanghavi].
- Designed continual learning methods using unsupervised latent partitioning for audio sensing, LoRA adapters, multi-sensor HAR, and cross-population glucose prediction; reduced labeling costs up to 3000× via dataset distillation.
- Developed multimodal/multisensory and cross-modal alignment techniques for time-series, language, health, and audio.

Bosch Research – Foundation Models Research Intern (BCAI) — Sunnyvale, CA
Mentors: Gorden Li, Wenbin He, Liu Ren — May 2026 – Aug. 2026

- Worked at the intersection of foundation models, SLM/LLM adaptation, and multimodal task-agnostic time-series Q&A.

Nokia Bell Labs – Research Intern (Device Intelligence team) — Cambridge, UK
Mentors: Soumyajit Chatterjee, Mohammad Malekzadeh, Fahim Kawsar — May 2025 – Aug. 2025

- Improved generalizability of PPG-based health foundation models using test-time adaptation.

Independent Research — Dec. 2018 – May 2023

- Analyzed the effect of calibration on prioritizing important samples during training [Mentor: Prof. Emtiyaz Khan, RIKEN].
- Worked on Bayesian active learning for on-device human activity recognition (HAR) and wearable sensing tasks.

Ericsson R&D, Global AI Accelerator (GAIA) – Data Scientist III — Chennai, India
Mentors: Shrihari Vasudevan, M J Prasath — Feb. 2019 – Apr. 2023

- Worked on ML for network intelligence and telecom, leading to multiple publications, patents, and AI-driven solutions.
- Advanced Ericsson's AI-Native telecom stack: 3GPP standardization for federated learning and multi-vendor model sharing, 5G NWDAF mobility and connectivity modeling, and unsupervised Bayesian anomaly detection (E-ADF).
- Built telco-AI platforms: Ericsson LLM Hub with domain-rich data and LLMs, language translation, active learning for customer-symptom modeling, and object detection for automated detection of infrastructure failures at cell sites.
- Designed zero-shot federated learning frameworks to handle new heterogeneous classes and models.

SmartCardia (EPFL) – Machine Learning Engineer — Chennai, India
Mentor: Srinivasan Murali — May 2018 – Nov. 2018

- Implemented gradient-boosted ensembles and LSTM models for regression and classification on imbalanced time-series wearable clinical data; engineered biomarker-driven features for sleep apnea, troponin, blood pressure, and hemoglobin.

Solarillion Foundation – Research Assistant — Chennai, India
Mentor: Vineeth Vijayaraghavan — Feb. 2016 – May 2018

- Led the development of HARNet – deep ensemble models for activity recognition with on-device incremental learning.
- Designed dynamic gesture recognition models with efficient feature engineering on a low-cost Raspberry Pi Zero ($5).

SSN College of Engineering – Undergraduate Student Researcher — Chennai, India
Mentors: Prof. Bhalaji N, Prof. Srinivasan R — Feb. 2015 – Mar. 2017

- Developed a CNN-based bus stop recognition system with ∼96% accuracy in dynamic Indian environments.
- Led an HCI research project on Neurocinematics to classify real-time cognitive responses of film viewers using EEG data.

Patents

- Efficient Backpropagation Free Test-Time Adaptation on Foundation Models for PPG *[Filed]*.
- Federated Learning using Heterogeneous Labels, WO2022013879A1.
- Distributed Machine Learning with New Labels using Heterogeneous Label Distribution, WO2022162677A1.
- Method and Apparatus for Approach Recommendation with Threshold Optimization in Unsupervised Anomaly Detection, WO2023166515A1.

Selected Talks

Resource-efficient Machine Learning — *SSN College of Engineering*, 2022
Heterogeneous Zero-Shot Federated Learning with New Classes for On-Device Audio Classification — *MobiUK*, 2021
Telecom-Specific Language Translation using GCP — *Ericsson/Google Cloud Day*, 2021
Resource-Constrained Machine Learning for Ubiquitous Computing Applications — *Flipped by GAIUS*, 2020

Selected Honors, Awards, and Fellowships

- SVFT was selected for an **oral presentation** at the WANT workshop at ICML 2024.

- **Graduate Ph.D. Fellowship** from the Cockrell School of Engineering at The University of Texas at Austin.
- **Top 1 percentile** on HackerRank in the Algorithms Domain – Problem Solving (Advanced).
- Full financial registration grants to attend ICLR 2021, NeurIPS 2020, and OxML 2020.
- Project *AIB* won Ericsson's **Top Performance Competition 2020** in the Operational Excellence category.
- Undergraduate research grant of INR 25,000 from SSN College of Engineering.
- **Winner** of the GermEval Shared Task 1 Challenge (Subtask (a)), KONVENS 2019, in the post-evaluation phase.

Professional Service

- **Organizing Committee:** WellComp 2026 workshop at ACM UbiComp/ISWC 2026.
- **Program Committee/Reviewer:** IEEE ICASSP 2026/2025, Interspeech 2026, IEEE MLSP 2026, ACM UbiComp/ISWC 2025, ACM ICMI 2025, ICML 2024 – ES-FoMo workshop, ICLR 2021 – DPML workshop, NeurIPS – ML4H 2020/2019.
- **Event Organizer:** Data Nuggets – a data science contest at Invente 2016, SSN College of Engineering.

Teaching and Mentoring

Teaching

- **Graduate Teaching Assistant, UT Austin** – Applied Machine Learning (Fall 2026), Deep Learning (Fall 2025), and Human Signals (Spring 2025) graduate courses. Primary instructors: Prof. Joydeep Ghosh, Prof. Edison Thomaz. Designed assignments, exams, and quizzes; mentored students on paper reviews, presentations, and projects.
- **Teaching Assistant, Solarillion Foundation** – Mentored 10+ students on embedded ML research projects and assignments (Feb. 2017 – May 2018).

Mentoring

- **Aadharsh Aadithya** (UT Austin) – Discrete diffusion models for time-series data (2026 – Present).
- **Ji Hwan Park** (UT Austin) – Audio sensing and multimodal alignment (2025 – 2026).
- **Abhijith Ragav** (GaTech → Amazon) – Bayesian active learning for wearable stress and affect detection (2020).
- **Prahalathan Sundaramoorthy** (USC → HP Labs) – Bayesian active learning for human activity recognition (2018).
- **Manav Rajiv Moorthy** (UWash → Amazon) – Deep learning and human activity recognition (2017 – 2018).

Summer Schools

- **5th Summer School on Artificial Intelligence** 2021, IIIT Hyderabad.
- **Eastern European Machine Learning Summer School (EEML)** 2020 and 2021. Presented ActiveHARNet (EEML '20); zero-shot federated learning and task-independent continual learning (EEML '21).
- **Oxford Machine Learning Summer School (OxML)** 2020.

Posters and Extended Abstracts

- Uncertainty-Aware Chain-of-Thought Reasoning in Large Language Models, *AI in Health* graduate course, UT Austin.
- Adaptive Federated Learning in Conceptually Drifting Environments, *Applied Machine Learning* graduate course, UT Austin.
- Heterogeneous Zero-Shot Federated Learning with New Classes for On-Device Audio Classification, MobiUK 2021.

- Bayesian Active Learning for Wearable and Mobile Health, NeurIPS Europe Meetup on Bayesian Deep Learning (BDL 2020).
- Handling Real-time Unlabeled Data in Activity Recognition using Deep Bayesian Active Learning and Data Programming, MobiUK 2019, University of Oxford.
- Neurocinematics: The Intelligent Review System, 3rd International Conference on Cognition, Brain and Computation (CBC 2015), IIT Gandhinagar.

Technical Coursework and MOOCs

- **Relevant Ph.D. Coursework:** Generative Models in Machine Learning, Advanced Computer Vision, Statistical Methods I & II, Ethics of AI, Applied Machine Learning, Fine-Tuning LLMs, AI in Health, Spoken Language Technologies, Human Signals.
- **HackerRank:** Problem Solving – Advanced, Intermediate, Basic.
- **University of Washington (Coursera):** Machine Learning Specialization (4 courses) – A Case Study Approach, Regression, Classification, Clustering & Retrieval.

Technical Skills

Programming Languages & Tools: Python, C++, HTML/CSS, Bash, SQL, JavaScript, Git, LaTeX, Claude Code, Docker, GCP

Libraries & Frameworks: PyTorch, TensorFlow, Hugging Face Transformers, Keras, NumPy, Pandas, Scikit-learn, PySpark